

Increasing Channel Occupancy in Large-Scale Mobile Radio Systems: Dynamic Channel REassignment

DONALD C. COX, SENIOR MEMBER, IEEE, AND DOUGLAS O. REUDINK, MEMBER, IEEE

Abstract—Large-scale mobile radio systems using dynamic channel REassignment switch the channels assigned to some of the calls in progress to maintain a nearly optimum separation between coverage areas simultaneously using the same channel. Dynamic channel REassignment produces a significant increase in channel occupancy. For example, in the systems studied, which have a uniform distribution of their fixed channels and were operated against a uniform spatial distribution of offered traffic, the channel occupancy was increased by $\frac{2}{3}$ over a pure fixed channel assignment system at a blocking rate of one percent. This corresponds to a channel savings of about 40 percent for the same traffic carried at one percent blocking by the hybrid systems that were studied.

I. INTRODUCTION

IF THE spatial distribution of the offered traffic to a large-scale multiple base station mobile radio system can be estimated accurately and does not change from hour to hour, then the traffic handling capability of the system can be increased by utilizing dynamic channel REassignment. Dynamic channel REassignment switches the channels assigned to some of the calls in progress so that the separation between coverage areas simultaneously using the same channel is more nearly optimum. The initial channel assignment used in this study makes use of coordinated dynamic channel assignment and fixed channel assignment techniques described previously [1]–[5]. This paper describes a computer simulation of large-scale two-dimensional mobile radio systems that use one form of dynamic channel reassignment and compares the performance of these systems with fixed channel assignment systems and dynamic channel assignment systems that do not employ channel reassignment.

II. THE SIMULATION MODEL

A. The Model of the System Configuration

The system simulated consists of a set of square radio coverage areas arranged to completely cover a square grid with no overlap as illustrated in Fig. 1. For the simulation, coverage squares were specified to be 2 mi on a side. Since traffic handling capabilities of dynamic channel assignment systems are the primary concern of this study, it is desirable to define geometric coverage areas that completely cover a region with no overlap. The geometry associated with squares is the

Y \ X	1	2	3	4	5	6	•	•
	A	B	C	D	A	B	C	•
1	A	B	C	D	A	B	C	•
2	E	F	G	H	E	F	G	•
3	I	J	K	L	I	J	•	•
4	M	N	O	P	M	N	•	•
5	A	B	C	D	A	B	•	•
•	E	F	•	•	•	•	•	•
•	•	•	•	•	•	•	•	•

Fig. 1. Coverage area grid with channel subsets as designated in Fig. 2.

simplest to manipulate. Since propagation in a mobile radio environment is highly irregular, radio channels obviously may not be used simultaneously in immediately adjacent coverage areas regardless of the geometry defining the coverage area. Thus, square coverage areas on a square grid provide a reasonable model for these traffic studies. Such a system could be approximated by placing base stations at the corner of coverage areas and radiating into them with directional antennas.

Channels allocated for possible use within coverage areas specified by *A* through *P* shown in Fig. 1, are indicated in Fig. 2. In Fig. 2, duplex radio channels are numbered 1 through *T* where *T* is the total number of channels available in the entire system. (For the simulation this number was specified to be 160.) In the example in Fig. 2 it is assumed that channels can be simultaneously used at a separation of four coverage areas. That is, the reuse interval is four. Some of the channels available to the system are assigned on a fixed basis. That is, certain channels can be used only at specified base stations. *F* represents the number of fixed channels allocated to each coverage area. (*F* may vary from 0 to *T* divided by reuse interval squared, in this case from 0 to 10.) The remaining channels [*T* - (reuse interval)² × *F*] are assumed to be available for assignment in any coverage area so long as they are not simultaneously used closer than a reuse interval apart.

B. The Stimulus to the Simulated System

The initiation of call attempts and the movement of vehicles in this simulation is almost identical with that described previously [2] for a one-dimensional simulation. Call attempts are generated in the simulation as a Poisson process in time that has a controllable average call attempt rate in each coverage area. For this study initial call attempts from vehicles are uniformly distributed in both the *X* and *Y* coordinates and the *X* and *Y* locations are independent. Call attempts are tapered at the edges of this simulated universe and the performance

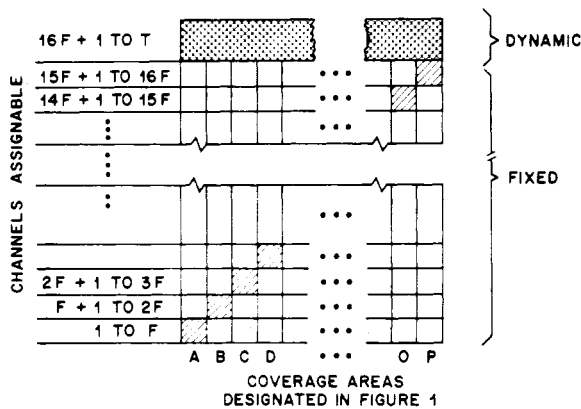


Fig. 2. Channel allocations for coverage areas.

data are accumulated only from the central portion of the system so that the results are representative of an infinite system.

In this method of simulation, vehicles making calls are identifiable entities whose locations and movements are stored in the computer. Vehicle velocity (speeds and direction) are chosen randomly from a population whose statistical characteristics can be prescribed. Vehicle motions are primarily parallel to one of the coordinate axis with 45 percent of the vehicles having an X velocity component only and 45 percent a Y component only. Ten percent of the vehicles have both X and Y velocity components that are mutually independent. The velocity components (X and Y) have a truncated Gaussian distribution with a zero mean, a standard deviation of 30 mi/h, and a maximum velocity magnitude of 60 mi/h [2, Fig. 8].

Call attempts that are assigned to a channel remain on for call durations that are taken from a random population with a specified distribution. The data to be presented later are for a call duration distribution that is exponential with a mean of 98 s.

C. The Simulated Dynamic Channel Reassignment Operating System

The method of handling call attempts, calls in progress, and call terminations is similar to that described in [2] for a one-dimensional system. The first step in processing a new call attempt is to assign a coverage area. Since coverage areas are geometrically defined and a vehicle's coordinates are known, the appropriate coverage area is readily determined. In the simulation vehicles within a particular coverage area are required to be served by channels available in that coverage area. The next step is a channel search that tries to find a radio channel to serve the awaiting call attempt.

A search is made first through the fixed channels allocated to the assigned coverage area. If there is a fixed channel not being used in the assigned coverage area, that channel is assigned to serve the call attempt. If all fixed channels are in use, a channel search is made through the dynamically assignable channels. The dynamic channel assignment procedure used in this study is the RING strategy discussed in [1]. If no dynamic channel is available, the call is immediately blocked and

cleared from the system. This blocking strategy is the same as that used in deriving the Erlang B (blocked calls cleared) telephone traffic formula.

Efficient use of channels requires the simultaneous assignment (reuse) of a radio channel in radio coverage areas that are spaced as close together as possible without incurring excessive cochannel interference. Dynamic channel reassignment is initiated when a caller using a fixed channel terminates his call. A search is made within the coverage area where the fixed channel call termination occurred to determine whether a dynamically assignable channel is being used within that coverage area. If a dynamically assignable channel is being used to serve a call, that call is transferred to the just vacated fixed channel. This frees the dynamic channel for future assignment and ensures that a large number of calls are being served by the optimally spaced fixed channels. Call terminations that occur on dynamically assigned channels are cleared from the system and no further action is taken.

The simulated system detects vehicles crossing coverage area boundaries and initiates a new channel search when this occurs. If no channel is available, the call is immediately forced to terminate and is cleared from the system. In an actual system the crossing of coverage area boundaries will not be critical and, hence, some delay could be tolerated in finding a new channel for boundary crossing vehicles. When a significant number of the calls handled by the system experience boundary crossings, the calls forced to terminate have an affect on other system performance parameters. With the coverage area size and the velocities used in this simulation, the effect of boundary crossings on the traffic carried and call blocking is small. Dynamic channel reassignment also is initiated when a caller using a fixed channel crosses a boundary and vacates the fixed channel.

D. System Parameters

The simulated operating systems were run for a 27 by 27 grid of coverage areas that were 2 mi on a side. The systems had available 160 duplex radio channels. With the assumed reuse interval of four this results in ten channels (160/16) available on the average at each base station. Of course, depending upon the system activity and the instantaneous demand, some coverage areas will have more than ten channels in use simultaneously and some will have fewer. The number of fixed channels allocated per coverage area (F in Fig. 2) was varied from 5 to 9 and correspondingly the number of dynamically assignable channels available on the average per coverage area (total dynamic channels available divided by reuse interval squared) varied from 5 to 1.

Because of the uniform spatial and temporal statistics of the call-attempt process, averages of system parameters taken over a large number of coverage areas at a particular time must yield the same results as averages taken over an extended time interval for one coverage area. Thus, the stability of the statistics for the performance parameters depends upon the product of the number of coverage areas included and the run time. Statistics were taken from the central 225 coverage areas (15

by 15) only to avoid any effects from the edges of the finite system. Call loading and system response at the edges of the finite system were tapered in a way that ensured that these central base stations were operating as members of an infinite set. The simulation was started initially with no calls on in the system and the statistics of parameters were monitored until the initial transients died out. The time required for stability was about 2 min. Data were collected after stabilization for about 7 min (in simulated time). This run time and spatial size was sufficient to produce data points that lie on smooth curves and have little scatter.

The data presented in the following sections were obtained by counting and storing the number of actual events, which occurred in the stabilized system. Some of the events counted were new call attempts, new call attempts blocked, coverage boundary crossings, and calls forced to terminate after crossing coverage boundaries. In addition, the actual number of calls in progress in the system was counted and stored periodically during each simulation run.

III. SYSTEM PERFORMANCE

System performance described in terms of traffic carried and the blocking of new call attempts as functions of traffic offered to the system is presented in Fig. 3. The traffic offered to the system is defined as the product of the new call attempt rate and the mean call duration and is expressed in the dimensionless quantity Erlangs (new calls per minute $\times$ minutes). The different curves are for the different mixtures of the number of fixed channels allocated per coverage area and the number of dynamic channels available on the average per coverage area as indicated in the figure key. The curves for the system with 10 dynamic channels available per coverage area and no fixed channels are for the two-dimensional dynamic channel assignment system (DYN SYS) described in detail in [1]. The curves drawn through the system with ten fixed channels and no dynamic channels (FIX SYS) were obtained from the Erlang *B* telephone traffic formulas [6]. The data points plotted along these FIX SYS curves were obtained by running the simulation with all of the vehicle velocities set equal to zero. The close fit of the simulation points to the theoretical curve provides a check on the accuracy of the simulation. All of the other curves are drawn through the data points from the simulation using velocities as described in Section II. The only effect of velocity is to shorten some of the call durations because of forced call terminations at coverage area boundaries. This effect will not change the curves since call duration is included in the parameters plotted.

The axis on the right-hand side of Fig. 3 and the curves to the lower right show the blocking of new call attempts as a function of the traffic offered. The blocking rate is the ratio of the number of new call attempts that are refused service within a given time period because of lack of channels to the total number of new call attempts in the same time period. The effectiveness of dynamic channel reassignment is illustrated by the fact that the blocking does not become significant (greater than 1 percent) until the traffic offered is over 50 percent greater than the traffic offered, which produces the same

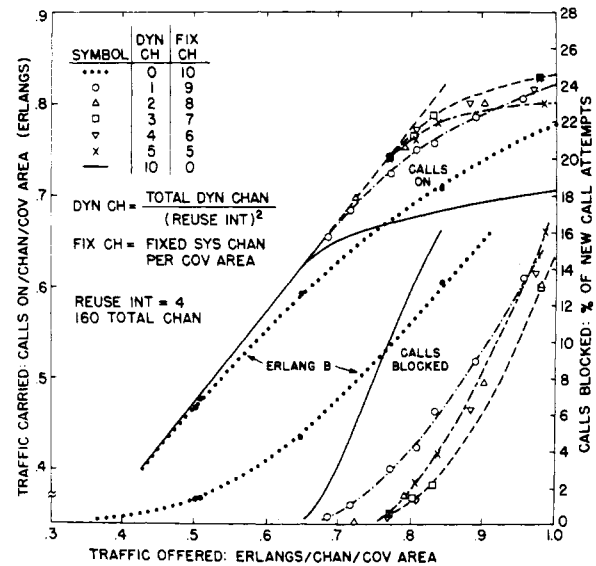


Fig. 3. Performance of dynamic channel REassignment systems.

blocking for the FIX SYS. It is also evident from the curves that the better performing dynamic channel reassignment systems do not block significantly until the traffic offered is about 20 percent greater than the traffic offered for which the DYN SYS produces the same blocking.

The left-hand scale and the associated curves of Fig. 3 show the relationship between the traffic carried and the traffic offered for the various systems. Traffic carried is expressed as the average number of calls on in the central coverage areas divided by that number of coverage areas (15 by 15) and is normalized to the total number of channels available on the average in each coverage area. It is obvious from the curves that all of the dynamic channel reassignment systems carry more traffic than either the FIX SYS or the DYN SYS within the region of good service (blocking rates less than 10 percent).

The curves in Fig. 4 show directly the relationship between the traffic carried and the blocking of new call attempts for the various systems. The performance of all of the dynamic channel reassignment system studied is far superior to the performance of either the FIX SYS or the DYN SYS. The reasons that the dynamic channel reassignment systems perform better are that 1) they have channels available that can be moved around to serve the normal random fluctuations in the offered traffic and 2) most of the channels are reused at the specified minimum channel reuse distance by reassigning whenever possible calls using dynamic channels to fixed channels. The systems that have from two to five dynamic channels available on the average per coverage area perform similarly and have nearly an exponential relationship between the traffic carried and the blocking rate over the range of operation studied. The system with nine fixed channels per coverage area apparently does not have a sufficient number of dynamic channels to meet the normal fluctuations of the offered traffic. Since the performance of the systems with five and nine fixed channels is somewhat inferior to those with six, seven, and eight fixed channels, there appears to be a broad optimum mixture of the fixed and dynamic channels approporionated to a system. This fact is clearly illustrated in Fig. 5 where the

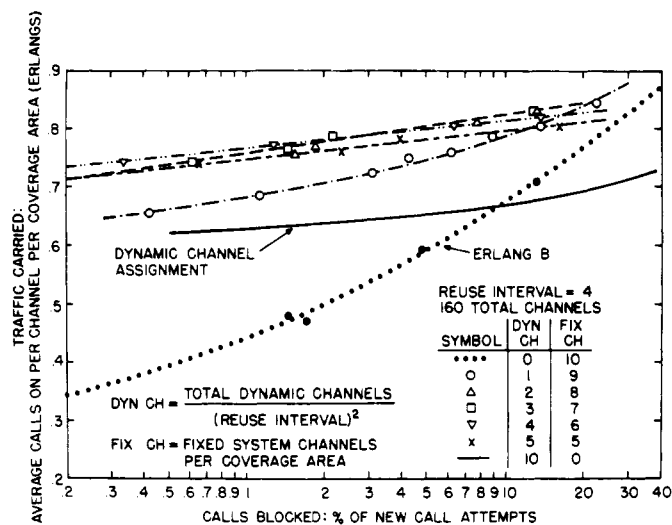


Fig. 4. Channel usage.

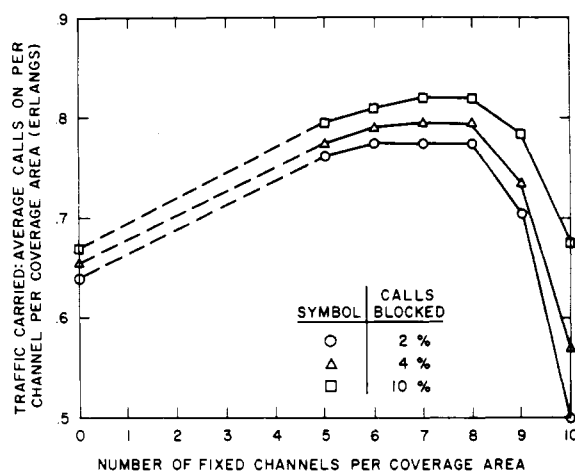


Fig. 5. Optimization of channel mix.

traffic carried is plotted versus the number of fixed channels per coverage area for three different blocking rates. Over a wide range of blocking, the system having only two dynamic channels available on the average can provide nearly optimum performance.

In addition to the overall system performance characteristics just presented, more detailed aspects of system operation are needed to assess the relative operating complexity of the various systems. Systems that have many dynamic channels allocated and set up many calls on dynamic channels are more complicated and therefore will be more expensive to build and operate. Such systems are obviously more flexible and less dependent upon accurate matching to the spatial distribution of offered traffic for efficient use of available channels.

The percentage of calls that are set up on dynamic channels is an indication of the computation complexity required to operate a system. Assigning calls to dynamic channels requires many more steps in computer processing than assigning calls to fixed channels. Also, rapidly tunable radio equipment required for dynamic channel assignment will probably be more expensive than fixed tuned radios. Fig. 6 shows the percent of call attempts that are set up on dynamic channels as a function of the call blocking rate for the various systems using the dif-

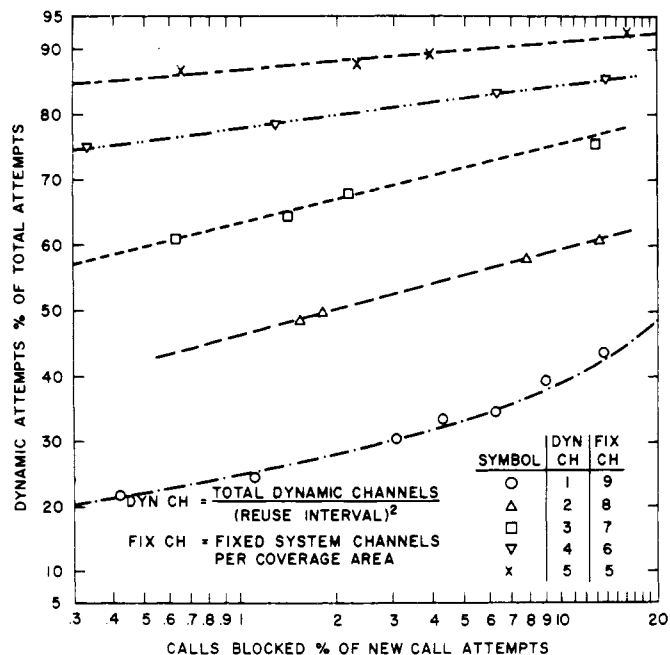


Fig. 6. Calls processed by dynamic channel assignment.

ferent mixtures of dynamic and fixed channels indicated. In order to achieve the high system performance illustrated in the previous figures, a large fraction of the calls are initially assigned to dynamic channels. For example, a system that has only 32 dynamic channels out of its total of 160 channels, sets up 50 percent of its calls on dynamic channels at a blocking rate of 2 percent. Obviously, at high blocking rates (> 25 percent) most of the calls will be set up on dynamic channels, even for systems with few dynamic channels.

Efficient channel reuse is achieved in these dynamic channel reassignment systems by switching calls initially assigned dynamic channels to fixed channels as soon as possible. This channel switching dramatically improves system performance at the expense of operating complexity. Although these systems switch channels of many calls in progress, this function must already be provided in the radio equipment to handle the channel switching required when coverage boundaries are crossed. Fig. 7 shows the percentage of total calls handled by the system that are switched from dynamic channels to fixed channels as a function of the blocking rate of new call attempts. As can be seen from the figure, this switching rate is not strongly dependent on the blocking rate. Again, to achieve the high overall system performance requires the switching of channels of a large fraction of the calls handled by the system. Since some of the calls cross coverage area boundaries and thus require more than one channel assignment, it is likely that a few calls will be switched from dynamic channels to fixed channels more than once while they are on in the system.

The channel efficiency of the fixed channels is increased considerably because of the switching of calls from dynamic channels into them. The dynamic channels efficiently handle the random fluctuations in offered traffic, which would otherwise degrade the performance of the rigidly structured fixed channels. Fig. 8 shows the resulting normalized channel efficiency of the fixed channels operating in the various dynamic

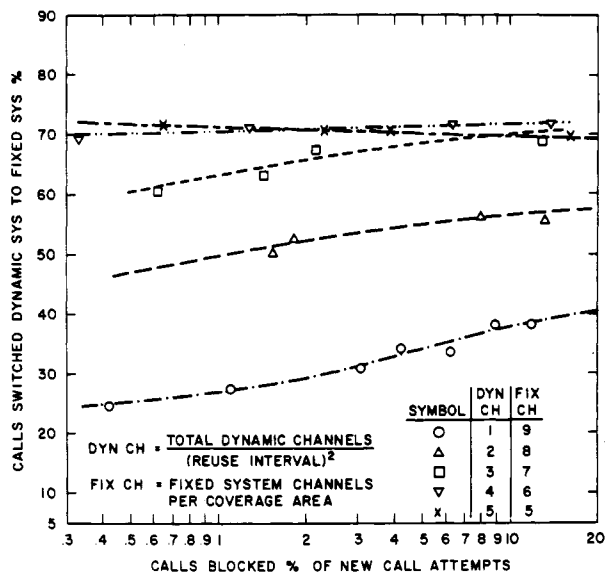


Fig. 7. Channel switching activity.

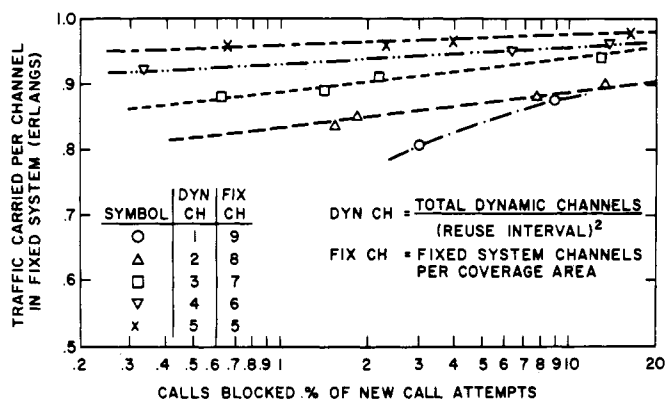


Fig. 8. Fixed assignment channel usage.

channel reassignment systems. A desirable feature of note is that the fixed channels in these systems operate at a high efficiency and the efficiency is relatively independent of the blocking rate. For example, at a blocking rate of 2 percent the channel reassignment system with 8 fixed channels and 2 dynamic channels available on the average in each coverage area utilizes its fixed channels with an efficiency of about 85 percent (the channel efficiency for the total system is 76 percent) whereas a FIXSYS with ten channels allocated per coverage area is only 50 percent efficient (see Fig. 4).

Another way of looking at the performance of these systems is to compare the channels required per coverage area for a FIXSYS to carry the same traffic as a 160 channel dynamic channel reassignment system using a specified mixture of fixed and dynamic channels. For example, Table I shows the number of fixed channels required to equal the performance of a dynamic channel reassignment system having eight fixed channels per coverage area and two dynamic channels available on the average per coverage area. At a blocking rate of 2

TABLE I
CHANNELS REQUIRED PER COVERAGE AREA FOR A FIXSYS TO CARRY THE SAME TRAFFIC AS A 160-CHANNEL DYNAMIC CHANNEL REASSIGNMENT SYSTEM USING EIGHT FIXED CHANNELS PER SQUARE COVERAGE AREA AT A REUSE INTERVAL OF FOUR

Blocking Probability	Number of Fixed Channels Required
1%	14+
2%	13-1/2
5%	12

percent a FIXSYS requires about one-third more channels to serve the same offered traffic as the hybrid system.

COMMENTS AND CONCLUSIONS

Dynamic channel REassignment in large-scale mobile radio systems produces a significant increase in channel occupancy. For example, in the systems studied that have a uniform distribution of their fixed channels and were worked against a uniform spatial distribution of offered traffic, the channel occupancy was increased by $\frac{2}{3}$ over a pure fixed channel assignment system at a blocking rate of one percent. This corresponds to a channel savings of about 40 percent for the traffic carried at one percent blocking by the hybrid systems studied.

The processing of new call attempts and the overall system control of hybrid systems are somewhat more complex than for fixed channel assignment systems. Hybrid systems with fewer dynamically assignable channels perform very well against matched loading, however, it is expected that the performance of these systems will be more seriously degraded by nonuniformities in offered traffic than systems with larger numbers of dynamically assignable channels.

REFERENCES

- [1] D. C. Cox and D. O. Reudink, "Dynamic channel assignment in two-dimensional large-scale mobile radio systems," *Bell Syst. Tech. J.*, pp. 1611-1629, Sept. 1972.
- [2] —, "Dynamic channel assignment in high capacity mobile communications system," *Bell Syst. Tech. J.*, pp. 1833-1857, July-Aug. 1971.
- [3] —, "A comparison of some channel assignment strategies in large-scale mobile communications systems," *IEEE Trans. Commun.*, vol. COM-20, pp. 190-195, Apr. 1972.
- [4] —, "The behavior of dynamic-channel-assignment mobile communications systems as a function of numbers of radio channels," *IEEE Trans. Commun.*, Part I (Concise Papers), vol. COM-20, pp. 471-479, June 1972.
- [5] —, "Effects of some nonuniform spatial demand profiles on mobile radio system performance," *IEEE Trans. Veh. Technol.*, vol. VT-21, pp. 62-67, May 1972.
- [6] *Reference Data for Radio Engineers* (ITT Handbook). Indianapolis, Ind.: Howard W. Sams & Co., 1968.

Donald C. Cox (S'58-M'61-SM'72), for a photograph and biography, see this issue, page 110.

Douglas O. Reudink (M'66), for a photograph and biography, see this issue, page 122.